

NOTE ANALYZE IMPROVE

12 chapters

~9 min

REFERENCES & SOURCES

ICAO Doc 9995 — Competency-Based Training & Assessment (CBTA)

ICAO Doc 9683 — Human Factors Training Manual (TEM)

EASA Part-FCL Appendix 9 — Type Rating Proficiency Check

EASA AMC/GM to Part-FCL — CBTA Grading Standards

BOEING NMS CBTA TEM Pilot Observer Document

BOEING 737 NG Quick Reference Handbook — Rev 55

BOEING 737 NG Flight Crew Training Manual — Rev 25

CRM FORDEC — Lufthansa Aviation Training · NITS Briefing Model

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CHAPTER 1 — 4-STEP QUICK START

1 SESSION — Configure pilots

On the home screen, enter the **staff number** of the CPT and FO — the name fills in automatically if the pilot is known. Assign roles: **CPT** / **FO** / **TRI CHECK**. Name the session (e.g. *B737 OPC DAAG Winter 2026*).

2 Select an exercise

Tap **List** then select your exercise from the library. Filter by flight phase: **TAKEOFF** **CLIMB** **CRUISE** **DESCENT** **APPROACH** **LANDING** · priority **CRITICAL** / **HIGH** / **STANDARD**, or type the name directly: e.g. *ENGINE FIRE*, *ILS CAT III*, *TCAS RA*.

3 During the exercise — Grade in real time

Grade **competencies** (1–5), add **+** observations (positive) and **-** (areas for improvement), identify **TEM threats** and check **FFA items**. Everything is done live — no interruption to the simulated flight.

4 Close → Automatic Debrief

Tap **CLOSE** to seal the data. Repeat for each exercise. At the end of the session, open the **DEBRIEF** — the CBTA radar, TEM, CRM and conclusion are generated automatically.

CONCRETE EXAMPLE

The CPT declares MAYDAY for engine fire during climb at FL150. You open the *ENGINE FIRE IN FLIGHT* exercise, activate competencies **PRO** **SA** **COM** and grade in real time.

CHAPTER 2 — CBTA GRADING SCALE

5 performance levels. Each grade corresponds to an observable defined behavior — not a subjective impression.

5

EXEMPLARY

Total mastery. Autonomous even under pressure. Can teach.

4

ABOVE STANDARD

Minor imperfections. Autonomous corrections. Very good level.

3

STANDARD ✓

Pass threshold. SOP/AOM compliant. Acceptable for line.

2

BELOW STANDARD

Visible gaps. Mandatory follow-up. Remediation to be planned.

1

UNACCEPTABLE — IMMEDIATE REMEDIATION

Dangerous or absent performance. Exercise must be repeated. Mandatory briefing before any progression.

⚠️ GOLDEN RULES

- A **3** = minimum pass threshold — never skip debriefing even a barely-passed 3.
- A **2** = remediation to be planned — NEVER leave without structured, documented feedback.

- A 5 = rare and exceptional — document the precise exemplary behavior in observations.

CONCRETE EXAMPLE

The PF spontaneously anticipates a 22 kt crosswind before even the approach briefing, adjusts Vref and briefs a GO criteria diversion. You note: (SA):4 (WLM):4 + obs+ "Proactive crosswind anticipation before ATC briefing".

CHAPTER 3 — THE 9 CBTA COMPETENCIES

KNOWLEDGE-BASED

(KNO) Knowledge — Application of Knowledge

B737 systems, limitations, EASA regulations — the theoretical foundation underpinning everything.

Ex: CPT hesitates on the VMO/MMO limit during rapid descent → KNO:2.

(PRO) Procedures — Application of Procedures

SOP, QRH, NNC, memory items — rigorous application in the correct sequence.

Ex: GEAR DN callout not performed at gear extension → PRO:2.

COGNITIVE

(SA) Situation Awareness

Continuous awareness of aircraft situation, weather, traffic, fuel.

Trigger: crew unaware that the alternate is CAVOK but closed.

(PSD) Problem Solving & Decision Making

FORDEC, adaptability, cognitive bias management — deciding quickly and correctly.

Ex: CPT wants to land despite 3 incomplete QRH items → PSD:1.

(WLM) Workload Management

Prioritisation, delegation, time management in complex situations.

Trigger: CPT manages checklist AND ATC AND ACAS simultaneously without delegating to FO.

BEHAVIORAL

LTW**Leadership & Teamwork**

Assertiveness, crew coordination, mutual cross-check.

Trigger: FO does not challenge an obvious CPT error — LTW:2.

FPM**Flight Path Mgmt Manual**

Trajectory and energy in manual flight — approach stability, deviation management.

Ex: unstabilised approach at 500ft — ILS deviation > 1 dot → FPM:2.

FPA**Flight Path Mgmt Automation**

AP, AT, FMC, LNAV, VNAV — programming, monitoring and taking back automatics.

Ex: FMC not reprogrammed after diversion → FPA:2.

COM**Communication**

ATC phraseology, crew exchanges, SOP callouts — clarity, conciseness, confirmation.

Trigger: PF responds to TCAS without reading the RA aloud — COM:2.

CHAPTER 4 — EXERCISE ANATOMY & LOFT**PRIORITY LEVELS****CRITICAL**

Vital EASA exercises — assessed every OPC/PC. Failure = blockage. Ex: ENGINE FIRE, EXPLOSIVE DECOMPRESSION, TCAS RA.

HIGH

Important exercises, recommended every cycle. Ex: SINGLE ENGINE APPROACH, WINDSHEAR ESCAPE, RNP AR.

STANDARD

Basic exercises, ongoing training. Ex: NDB APPROACH, VOR/DME, ILS MANUAL, PATTERN WORK.

FLIGHT PHASES — AVAILABLE FILTERS**TAKEOFF****CLIMB****CRUISE****DESCENT****APPROACH****LANDING****LOFT — LINE ORIENTED FLIGHT TRAINING****Complete airline flight simulation**

LOFT is an immersive scenario simulating a realistic airline flight from A to B — with authentic ATC, weather, diversions, progressive failures and cabin interactions. No interruption during the session: the crew manages everything as in real operations.

- ✓ Assesses all 9 CBTA competencies in an integrated, realistic context
- ✓ Requires continuous grading throughout the flight — activates multiple successive exercises
- ✓ CRM and TEM evaluated in their natural context — no artificial scenarios
- ✓ Ideal for OPC/PC at end of qualification and annual reviews

LOFT STRATEGY IN THE APP

1. Open exercises **in flight order** (**TAKEOFF** → **CLIMB** → **CRUISE** → **DESCENT** → **APPROACH** → **LANDING**)
2. Close each exercise **as soon as it is complete** — without reopening the debrief in between
3. Inject **random threats** to simulate unexpected ATC or systems events
4. At LOFT end → open Debrief → the CBTA radar rebuilds the full session profile

EXERCISE CARD ANATOMY

ENGINE FIRE IN FLIGHT

NNC · QRH MAN.31 · 12 min

CRITICAL

CLIMB

CRUISE

APPROACH

PRO

SA

PSD

COM

⚠ MANDATORY EXERCISES — EASA REQUIRED

MANDATORY exercises are the regulatory items that every OPC or PC must contain. The app displays them in a **dedicated colored bar** at the top of the exercise list for instant access.

B737 MANDATORY EXERCISES — OPC/PC EASA

- RTO (Rejected Takeoff)
- SE Go-Around
- Non-Precision Approach
- Windshear Escape
- SE ILS Approach
- SE Landing
- TCAS RA
- Upset Recovery

T / M BADGES ON EACH EXERCISE

T

Training — free training exercise, no regulatory constraint

M

Mandatory — EASA required · minimum grade 3 required · 2nd attempt if failed

🔄 2nd ATTEMPT MANDATORY — FULL PROCEDURE

- 1 The MANDATORY exercise is closed with a grade **< 3** → a **2nd** badge and a **🔄 2nd ATTEMPT** button appear automatically in the list.
- 2 Tap **🔄 2nd ATTEMPT** → a duplicate of the exercise is created with the *2nd* suffix. Perform the exercise a second time with the crew.
- 3 Close the 2nd attempt → the 2nd attempt score appears in the report. Both attempts appear in the Exercises tab for full traceability.

⚠ **EASA required:** if the 2nd attempt is also **< 3**, record in instructor notes + mandatory remediation briefing before line progression.

CHAPTER 5 — SCORING DURING THE SESSION

Real-time scoring is the heart of the app. Here are the 5 sub-steps in order.

1 Tap the exercise — it opens

The exercise timer starts automatically. You see the 4 tabs: EVAL · FFA · TEM · FACILITATION. The exercise name and QRH page are displayed at the top.

2 EVAL tab — grade 1 to 5

For each active competency (pre-selected for the exercise), tap the 1 to 5 button. Switch between CPT and FO to grade each pilot separately. Score color updates in real time.

3 Observations: + positive / – negative

+ OBS button for a positive behavior. – OBS button for an area for improvement. Type precise text — e.g. "Announced GEAR DOWN before outer marker without prompt". The more precise, the more valuable the debrief.

4 FFA tab — check FLY / FOCUS / ACT

Three checkboxes: FLY (trajectory OK?) FOCUS (situational awareness OK?) ACT (appropriate actions?). Checked = YES · Unchecked = NO to debrief.

5 TEM tab — identify threats + errors

Add external threats, crew errors and detected UAS. Each item is automatically timestamped. See Chapter 6 for full TEM details.

CONCRETE EXAMPLE

You observe that the PF does not sequence ECAM items after a configuration warning: open the EVAL tab → PRO → grade 2 → then obs– "CONFIG ECAM items not sequenced after alert — QRH sequence not followed".

CHAPTER 6 — TEM: THREATS & ERRORS

The ICAO TEM (Threat & Error Management) model structures risk analysis into 4 categories. You record them live in the TEM tab of each exercise.

▲ THREATS — External to the crew

Events or conditions imposed from outside. The crew does not cause them but must manage them.

B737 examples: 25 kt crosswind not briefed · late ATC instruction · system failure · runway contamination · ACAS traffic.

● ERRORS — Committed by the crew

Crew actions or inactions that deviate from SOPs or intentions.

Examples: GEAR callout missed at 1000ft · wrong altimeter setting · QRH not completed before approach · FCU mis-programmed.

◆ UAS — Undesired Aircraft States

Undesired aircraft states resulting from uncontrolled threats or errors.

Examples: heading deviation > 15° · speed out of range · altitude not held ± 200ft · unstabilised approach.

✓ RECOVERIES — Successful corrections

Crew actions that bring the aircraft back to a desired state after a UAS or detected error.

Examples: go-around at the right moment · spontaneous heading correction before limit · FO challenge on altitude.

COUNTERMEASURE LEVEL PER ITEM

OK ✓

Threat/error identified AND correctly managed

PARTIAL ~

Managed incompletely or late — to be debriefed

NO ✕

Not identified or not addressed — critical point to address

CONCRETE EXAMPLE

Threat: "25 kt crosswind → identified but approach briefing incomplete" → PARTIAL

Error: "GEAR DOWN callout missed at 1000ft radar altitude" → NO

UAS: "Heading deviation > 15° during QRH procedure"

Recovery: "FO announces heading — CPT corrects spontaneously before limit" → OK

CHAPTER 7 — FFA: FLY FOCUS ACT

The FFA model is the ICAO situational awareness filter. Three hierarchical levels — if the first fails, the others become meaningless.



FLY

Trajectory control & parameters

Is the aircraft safe? Altitude, speed, trajectory, energy — mastery of pure flying. If FLY is NO, everything stops there.



FOCUS

Situational awareness + anticipation

Is the crew managing the right priority? Are they aware of the main threat? Do they anticipate the next 5 minutes?
FOCUS NO = tunnel vision or saturation.



ACT

Decision making + appropriate actions

Are actions appropriate, sequenced, and proportionate to the situation? ACT NO = inappropriate decision or lack of critical initiative.

QUICK INTERPRETATION

FLY+FOCUS+ACT Perfect sequence — flight mastered end to end ✓

FLY OK / FOCUS NO Pilot in reactive mode — no anticipation to work on

FLY NO Loss of basic control — absolute remediation priority

CONCRETE EXAMPLE

The PF deviates 200ft on a CAT II ILS final while managing an ECAM alert: **FLY:NO** (deviation), **FOCUS:OK** (aware of approach), **ACT:OK** (corrects quickly). In debrief you ask: "What distracted you from trajectory control?"

CHAPTER 8 — CRM: CREW RESOURCE MANAGEMENT

CRM (Crew Resource Management) is graded in the **DEBRIEF → CRM** tab, not in the exercise. It evaluates the entire session on 12 EASA NOTECHS behavioral items.

- 1 Complete and relevant briefing (departure, approach, abnormal situations)
- 2 Standard SOP callouts consistently respected
- 3 Challenge & Response — systematic checklist read and response
- 4 Clear, concise and confirmed communication between pilots
- 5 Situational leadership — assertive without being authoritarian
- 6 Effective mutual cross-check — cross-monitoring of parameters
- 7 Workload management — delegation and prioritisation of tasks
- 8 Shared Situation Awareness — same mental picture in both seats
- 9 Adapted decision making — FORDEC method applied under pressure
- 10 Appropriate intervention on deviation — FO challenges CPT if necessary
- 11 Automation management — AP/AT monitored and taken back if necessary
- 12 Sterile cockpit respected below 10,000ft — no non-operational conversations

EASA NOTECHS LEVELS — PASS MARK: 70% ($\geq 9/12$)

$\geq 85\%$ EXCELLENT Exemplary CRM — very effective crew

70-84% SATISF. Regulatory level reached

50-69% PARTIAL Behavioral gaps identified

$<50\%$ INSUFFIC. Mandatory CRM remediation before line

⚠ IMPORTANT REMINDER

CRM is graded in the **DEBRIEF** → **CRM** tab — not in each exercise. It is a global evaluation of the entire session.

CHAPTER 9 — THE DEBRIEF — FULL ANALYSIS

6 tabs to dissect performance. Each tab is a different view of the same data.

SCORES — CBTA Radar

Spider web radar with **CPT circle (●)** and **FO diamond (◆)** overlaid to compare profiles. CBTA-colored competency score bars. Overall score displayed large.

READING THE RADAR

- The larger and more centered the polygon → the better the overall performance.
- A dip toward **FPM** = identified flying improvement point.
- CPT (●) vs FO (◆) overlaid → compare both pilots' profiles.

EXERCISES — Detailed summary

Each closed exercise listed with: color-coded average score, timed duration, T/M indicator, and direct access to recorded observations. You can reopen the detailed card of any exercise.

TEM — Global summary

All threats, errors and UAS from the entire session grouped together. Stats: number of threats mitigated / errors / UAS / successful recoveries. Ideal for identifying recurring patterns across multiple exercises.

CRM — Behavioral assessment

Visual donut with CRM % + level badge + list of 12 OK/NOK items separated for CPT and FO. EASA/NOTECHS pass mark: 70%. Reference EASA ORO.FC.115 and NOTECHS framework.

QUESTIONNAIRE — Trainee self-assessment

Reflection questions addressed to the trainee to gather their own perception of the session. Essential for comparing the instructor's view and the pilot's self-awareness.

CONCLUSION — Auto-generated summary

Automatic session summary with individual CPT and FO recommendations targeting weak competencies. Free instructor notes area. Professional PDF export in one tap.

PROFESSIONAL PDF REPORT

↓ PDF button at the top of the Conclusion → professional report generated in one tap, ready for archiving.

- **Page 1** — Session identity, CBTA scores, CPT+FO spider web radar
- **Page 2** — Detailed exercise-by-exercise summary with scores, durations, observations
- **Page 3** — TEM summary + CRM assessment + behavioral score donut
- **Page 4** — Individual CPT and FO recommendations + instructor notes

DATA SHOW — Briefing room projection

DS → full screen optimised for projector or large display. Giant CBTA radar + score bars readable from the back of the room. Switch between tabs in full screen for group debrief.

REF — TEM Card & Core Observables Boeing

REF → **TEM CARD** tab (Boeing threat/error/UAS definitions) + **CORE OBS** tab (9 blocks per competency, observable behaviors listed).

OPTIMAL GROUP DEBRIEF WORKFLOW

- ① **DS** → Show scores to crew on large screen — let them react
- ② **REF → CORE OBS** → Tap weak competency → pilots position themselves in listed behaviors
- ③ **↓ PDF** → Generate report → archive in pilot file

CHAPTER 10 — PRO TIPS & SHORTCUTS

Light / Dark Mode

Button  at the top right of any screen. Light mode for bright briefing room, dark mode for low-light simulator. Setting is remembered.

Pilot history & progression curve

Dashboard → PILOTS tab → tap a name → **Progression** button. You see the multi-session curve for each competency + cumulative XP bars. Powerful for annual reviews.



Session timer — start BEFORE the 1st exercise

The SESSION timer does not start automatically. Tap  when entering the cabin. The legal limit is 4h00 — alerts at 2h, 3h50 and 4h00.

Random threat — TEM injector

In an open exercise →  button → 16 scenarios in 5 categories, or fully random injection. The threat inserts into CPT+FO TEM with automatic timestamp. Ideal for surprising the crew and evaluating **WLM** and **SA** live.

 **ATC (4)** — Diversion · NORDO 7600 · TCAS RA · Runway blocked

CABIN (4) — Medical · Door unarmed · Smoke · Disruptive passenger

 **WEATHER (3)** — Alternate unavailable · Windshear · Unexpected icing

 **RUNWAY (1)** — Contamination / Braking action POOR

 **SYSTEM (4)** — Fuel imbalance · Pressurisation · Cargo fire · Bird strike

Double-tap a score — clear and restart

Graded too quickly? Double-tap the already-selected score button to deselect it and reset the competency to zero. No complex manipulation — fast and discreet during the flight.

PWA — Works offline

The app works **without internet** from the first visit with wifi. Ideal in the simulator. All data remains local.

BONUS — REF → CORE OBS

REF button in the debrief bar → **CORE OBs** tab. Complete directory of CBTA behaviors per competency. Ideal during debrief for trainees to self-identify in the listed behaviors. Cross-references 9 competencies with Boeing observable behaviors.

CHAPTER 11 — FAIR GRADING

An incorrect grade — through sympathy, habit or lack of knowledge — distorts the CBTA assessment and can **favour or penalise a pilot**. This chapter gives you the tools to grade objectively.

THE 1-5 SCALE — WHAT EACH GRADE MEANS

- 1 UNACCEPTABLE**
Dangerous behavior or total absence of the competency. Compromises flight safety. Requires immediate instructor intervention.
- 2 BELOW STANDARD**
Partial competency. Notable gaps persist. The pilot needs improvement before operational validation.
- 3 STANDARD**
Expected level. The pilot demonstrates the competency consistently and without intervention. This is **the norm, not mediocrity**.
- 4 ABOVE STANDARD**
Clearly superior performance. The pilot anticipates, shares, adapts. Exemplary behaviors on multiple aspects of the competency.
- 5 EXEMPLARY**
Exceptional and consistent mastery. The pilot elevates the crew's level. Reserved for truly remarkable performances — **rare**.

THE 5 BIASES THAT DISTORT A GRADE

HALO EFFECT

A likeable or reputedly good pilot receives 4s everywhere even if a competency was insufficient that day. **Grade what you SAW, not what you think of them.**

RECENCY BIAS

You only grade the last 5 minutes of the exercise. A good landing doesn't compensate 20 minutes of degraded WLM. **Grade over the full duration.**

SYSTEMATIC LENIENCY

Giving 3 by default to avoid conflict or not discourage. A 3 must be earned — if you have not observed the standard, put 2. **Kindness does not protect the pilot.**

EXCESSIVE SEVERITY

Never putting 4 or 5 "to avoid making them arrogant". A well-deserved 4 motivates and calibrates the pilot. **Refusing to grade high is also an error.**

CONTRAST BIAS

You just assessed a very poor pilot — the next one seems excellent by comparison. Each pilot is graded **against the absolute CBTA reference, not against others.**

THE OBSERVABLE RULE

An observable must be **seen, heard or measured**. Never an interpretation or a feeling.

✗ "Poor situational awareness"

✓ "PM did not announce FL100 crossing — 200ft deviation not verbalised for 40 seconds"

✗ "He can't make decisions"

✓ "Decision to continue approach taken unilaterally — diversion option never raised, FORDEC not applied"

HOW TO GRADE IN THE APP — STEP BY STEP

- 1 Open the exercise → **EVALUATION** tab
- 2 Select **CPT** or **FO** via the switcher at the top
- 3 Tap the relevant competency (e.g. **PSD**) → the grading panel opens
- 4 Tap the grade **1 to 5** — the grade color lights up
- 5 Tap **+ OBS** to add a positive observable or **- OBS** for negative
- 6 Write the observable as **observed behavior** — what you saw or heard, not your interpretation
- 7 Switch to **FO** → repeat steps 3–6 for the co-pilot

SUGGESTED GRADING — HOW IT WORKS

After checking observables, the app displays a **suggested grade in orange**. This is not an automatic decision — it is a starting point for your reflection. **You are the only one to judge.**

TWO MODES DEPENDING ON HOW YOU GRADE:

MODE 1 — You only note deficiencies

Fast workflow: you only check what went **wrong** (the "-"). Unchecked behaviors are considered **acceptable**.

FORMULA

$$\text{score} = (\text{total official OB} - \text{nb of "-"}) \div \text{total OB}$$

Example — PSD competency (9 official OBs):

1 deficiency $(9-1) \div 9 = 89\%$

4 Above

3 deficiencies $(9-3) \div 9 = 67\%$

4 Above

5 deficiencies $(9-5) \div 9 = 44\%$

2 Below standard

9 deficiencies $(9-9) \div 9 = 0\%$

1 Unacceptable

MODE 2 — You note good AND bad behaviors

Detailed workflow: you check "+" and "-". The calculation is based solely on **what you actively noted**.

FORMULA

$\text{score} = \text{nb of "+"} \div (\text{nb of "+"} + \text{nb of "-"})$

Example — PSD competency:

7+ and 1- $7 \div 8 = 87\%$

4 Above

5+ and 3- $5 \div 8 = 62\%$

4 Above

3+ and 5- $3 \div 8 = 37\%$

2 Below standard

CONVERSION TABLE (% → GRADE)

≥ 90%	5	EXEMPLARY
≥ 70%	4	ABOVE STANDARD
≥ 50%	3	STANDARD
≥ 30%	2	BELOW STANDARD
< 30%	1	UNACCEPTABLE

THE INSTRUCTOR IS ALWAYS MASTER OF GRADING

The suggested grade is an **aid tool**, not a decision. The algorithm counts checkmarks — it does not see the situation, the pilot's energy, the scenario complexity, nor the CRM context. **You do.**

- ✗ Never tap "APPLY" without thinking — read the suggestion, compare to your overall impression.
- ✓ Use the suggestion as an **anchor point**: if it says 4 and you were hesitating between 3 and 4, it confirms 4. If it says 3 but the pilot was clearly struggling, put 2.
- ✓ A single **dangerous** behavior can justify a 1 even if other observables were positive — safety takes precedence over the average.

FULL EXAMPLE — SE ILS (SINGLE ENGINE ILS)

Exercise: engine failure after V1, single-engine ILS approach, go-around

KNO 3 ENG FIRE procedure correct — 1 item missing (FUEL CROSS FEED)

WLM 4 Load well distributed — PM managed radio while PF flew

SA 3 Situational awareness maintained — destination weather not checked

PSD 2 FORDEC not applied — go-around decision taken unilaterally without consulting PM

LTW 3 Correct coordination — PM not briefed before GA (no NITS)

FPM 4 ILS track and glide path precise throughout — deviations < ½ dot

DEBRIEF READING

The 3.2 average reflects a technically solid pilot but with a PSD/LTW deficit. The debrief focuses on **FORDEC and NITS** — not on technique which is at standard.

CHAPTER 12 — OFFLINE MODE

The app works **100% offline** after the first visit with a connection — ideal for simulator environments where wifi is unavailable or unreliable.

ACTIVATION — ONE TIME ONLY

1 Open the app once with a connection

Visit the GitHub Pages URL once on Safari (iPhone) or Chrome (Android). The app caches automatically in the background — this is silent and automatic.

IN THE SIMULATOR — NO CONNECTION NEEDED

- ✓ Turn off wifi or enable **Airplane Mode** — the app opens exactly as normal
- ✓ All features available: exercises, grading, debrief, guide
- ✓ Session data saved locally on device — never lost
- ✓ PDF generation and export work fully offline

TEST BEFORE THE EXAM DAY

1. Open app with wifi
 2. Disable wifi
 3. Close and reopen the app
- If it loads = offline operational ✓

AUTOMATIC UPDATES

When a new version is published, the app updates silently in the background on the next connection. No action required.

QUICK REFERENCE

CBTA GRADES — COLOR CODE

5

EXEMPLARY

Full mastery — fully autonomous

4

ABOVE STANDARD

Good performance, minor corrections

3

REQUIRED STANDARD

Acceptable, AOM compliant

2

BELOW STANDARD

Deficiencies — monitoring required

1

UNACCEPTABLE

Unacceptable performance — remediation required

THE 9 CBTA COMPETENCIES

FPM Flight Path Management — trajectory & energy *manual*

FPA Flight Path Automation — AP / AT / FMC / LNAV / VNAV

PRO Procedures — SOP, QRH, NNC, memory actions

SA Situation Awareness — continuous awareness of the situation

WLM Workload Management — prioritisation, task delegation

COM Communication — ATC phraseology, crew exchanges, callouts

LTW Leadership & Teamwork — assertiveness, coordination, cross-monitoring

PSD Problem Solving & Decision — FORDEC, adaptability, cognitive biases

KNO Knowledge — systems, limitations, EASA regulations

EXPERT LEVEL BADGE

Some exercises are marked **EXPERT** — advanced procedures requiring multi-competency mastery. Ex: ILS RAW DATA, AIRSPEED UNRELIABLE, RNP AR...

WORKFLOW — 8 STEPS

1 WELCOME — Instructor Station

- ① Enter CPT and F/O **staff number** → name auto-fills if known.
- ② Roles: **CPT** / **FO** / **TRI CHECK**.
- ③ Name the session (e.g.: *B737 OPC LVO*).
- ④ Optional:  button to load a **Loft**.

2 PRE-SESSION BRIEFING

Check pre-flight items, review EASA tolerances, then  **START**.

3 SESSION — Add exercises

 **EXERCISE** → search by name, competency or phase (**TAKEOFF CLIMB CRUISE DESCENT APPROACH LANDING**).
MANDATORY bar at top = EASA mandatory items.
Start the **SESSION timer** before the first exercise.

4

EXERCISE — 4 assessment tabs**EVALUATION** — Grades 1–5 per competency + observables**FFA** — FLY / FOCUS / ACT**TEM** — Threats · Errors · UAS · Countermeasures**FACILITATION** — Debrief questions

5

T / M — Training or Mandatory• **T** = free training · • **M** = EASA mandatory.Failure (<3) → 🔄 **2nd ATTEMPT** button appears automatically.Done → **CLOSE**.

6

DEBRIEF — 6 tabs**Scores** — CBTA Radar CPT+FO + competency bars**Exercises** — Summary of each closed exercise**TEM** — Threats / errors / UAS synthesis**CRM** — 12 items · Pass mark 70%**Questionnaire** — Trainee self-assessment**Conclusion** — Summary + recommendations + PDF

7

REPORT — PDF & Email**REPORT** button → full report with scores + TEM + conclusion.✉ **EMAIL** button for direct sending. Archived in history.

8

END — Save & History**END** button → session archived with scores, PDF report, TEM data. Find in **Dashboard** → **History**.**EXERCISE LIBRARY****MEMORY**

Immediate memory actions — no checklist

MANEUVER

Special maneuvers — TCAS, Windshear, GPWS, Upset...

FLIGHT PATTERN

Approach procedures and standard patterns

NNC

Non-Normal Checklists — system failures QRH

MANDATORY

EASA mandatory — RTO, SE ILS, SE GA, SE LDG, TCAS, Windshear, Upset

LVO

Low Visibility Operations — CAT II/III, LVTO, LVTO REJECT

RNP

RNP and RNP AR approaches — precision navigation

SMART SEARCH

Type a name (*ENGINE FIRE*), competency (*SA, WLM*), QRH page (*MAN.1*), or phase (*TAKEOFF*). Category and phase filters in real time.

SESSION LOFTS

What is a Loft?

A saved, reusable exercise list. Ex: create *OPC Standard 2026* once, reload it in one tap every session.

Create a Loft

 button → + **NEW LOFT** → name it → + **Add** → select exercises → **SAVE**.

Load a Loft

Before session: ▶ Load → exercises added at startup.

Active session: exercises added immediately.

LOFT IDEAS

OPC Standard · LVO Check · Initial Type Rating · Pilot Remediation

PILOT IDENTIFICATION — STAFF NUMBER

- ① Enter the company staff number in the dedicated field (e.g.: *CA-1234*).
- ② If the pilot is known, the badge changes from **NEW** to  **N SESSIONS** and the name fills in automatically.
- ③ If a new pilot, enter the name manually — it will be remembered next time.

WHY THE STAFF NUMBER?

Two pilots may have the same name. The staff number ensures the correct progression file is retrieved, even months later or by another TRI.

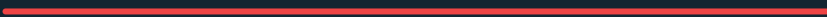
⌵ TIMERS — SESSION · EXERCISE · TIME LIMIT

① SESSION TIMER — HH:MM:SS (manual)

Displayed top left. **Does not start automatically.** Press  when ready.

0 → 2h00  **GREEN** Normal

2h → 3h50  **ORANGE** beep 2h00

3h50 → +  **RED** flashing +

4h00 → LEGAL LIMIT

② EXERCISE TIMER — MM:SS (automatic)

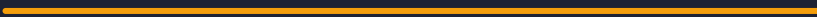
Starts automatically when exercise opens. ■ PAUSE and ■ STOP buttons.

Colors: white (running) · orange (paused) · grey (stopped).

③ TIME LIMIT — Configurable clock

Click -- : -- on the exercise → cockpit digital clock. Adjust -10/-5/-1/+1/+5/+10 min. Customisable neon colour.

< 80% limit  **GREEN**

80% → 100%  **ORANGE** — pulsing

≥ 100%  **RED** — flash + halo

⚡ S / L — SAVING OR LOSS

When closing an exercise with a time limit: badge **S:Xm** (green = time saved) or **L:Xm** (red = time exceeded).

Summary in Debrief → Exercises.

⚡ TIME MANAGEMENT

Per-exercise time tracking — visible in Debrief → Exercises

① ACTIVATE A TIME LIMIT

In the exercise list, tap the  button on a card → enter duration in minutes → confirm. The exercise timer starts automatically when the exercise opens.

② SAVING / LOSS AT CLOSE

When the exercise closes, a badge appears on the card:

S:Xm green = crew finished **before** the limit

L:Xm red = crew **exceeded** the limit

③ RECAP IN DEBRIEF

Debrief → **Exercises** tab → TIME MANAGEMENT block:

- Table per exercise: set limit + S or L in minutes
- **Average per exercise** — overall time management indicator
- If loss detected → automatic **recommendations** (briefing, checklist, coordination)

⚠ IF NO TIME LIMIT SET

The TIME MANAGEMENT block remains visible with an invitation message. Activate at least one time limit on an exercise to see the data.

INSIDE AN EXERCISE — THE 4 TABS IN DETAIL

EVALUATION — Grades & Observables

Grades 1–5 per competency. Observables  / . Switch **CPT** ↔ **FO**.

FFA — FLY · FOCUS · ACT

FLY — Aircraft secured? Position, energy, trajectory.

FOCUS — Crew managing the primary threat?

ACT — Actions adapted and prioritised?

TEM — Threat & Error Management

▲ **THREATS** — external events (weather, ATC, failure...)

● **ERRORS** — crew actions/inactions

◆ **UAS** — Undesired Aircraft States

Countermeasure: ✓ **Managed** / ~ **Partial** / ✗ **Unmanaged**. Auto timestamp.

FACILITATION — Debrief questions

Prepared questions to guide the trainee's reflection. Check off those addressed during the debrief.

TRAINING MODE — GHOST SESSION

Button at the bottom of the welcome screen. Training mode sessions are **NOT saved** in history. Ideal for preparing or testing a new scenario.

TRAINING

Cyan = inactive (normal mode)

TRAINING

Orange = active (ghost session)

TRAINING MODE

banner at top of session — impossible to confuse with a real session.

BACKUP — JSON EXPORT & IMPORT

Export

History → **↓ Export** → JSON with all sessions and pilot files. Save to iCloud / Google Drive.

Import

↑ Import → **MERGE** mode (adds without overwriting) or **REPLACE** mode.

External session

button → import a report from another TRI. **IMPORTED** badge in history.

IMPORTANT

Cache cleared = data lost. **Export after every session.**

HISTORY & PILOT FILE

DASHBOARD button on the welcome screen. 3 tabs:

DASHBOARD — Global KPIs, fleet competencies, latest sessions

SESSIONS — Chronological list, access to PDF reports

PILOTS — Average score, trend, XP bars, Progression button

VIEW FULL FILE → all pilot sessions + PDF reports.

Progression → multi-session curve + cumulative XP bars.

XP SYSTEM — CUMULATIVE PROGRESSION

XP = sum of grades across *all* sessions. A pilot with 15 XP in KNO has demonstrated their knowledge across multiple sessions.

ROOKIE

0–1 XP

STANDARD

2–5 XP

CONFIRMED

6–11 XP

ADVANCED

12–19 XP

EXPERT

20+ XP — multi-session mastery demonstrated

XP bars visible in each pilot file (Pilots tab) + Progression overlay.

QUICK TIPS

- ▶ **SESSION timer** — start it *before* the first exercise, not after
-
- ✓ Always **CLOSE** an exercise for it to count in the debrief and history
-
- ⚠ **CRM** = tab in the **DEBRIEF**, not in the exercise
-
- Random threat** — inject without warning to test WLM/SA in real conditions
-
- REF → CORE OBs** — use during debrief so trainees self-identify behaviors
-
- Always enter the **staff number** to link the session to the correct pilot file
-
- 🔍 Font size: **A- / A / A+ / A++** buttons at the top right of the welcome screen
-
- ☀ / 🌙 Day/night mode — button available on all screens
-
- 📄 **Export JSON after every session** — it is your only external backup
-

THREAT INJECTOR — HOW TO USE

 button in the bottom bar of an open exercise. Two modes:

RANDOM INJECTION

Tap **RANDOM INJECTION** to surprise the crew with a random threat. The threat inserts **simultaneously** into **CPT** and **FO TEM** with automatic timestamp. No warning to trainees — ideal for testing WLM/SA reactivity.

CHOOSE FROM LIST

Select a specific threat for a targeted exercise. Each threat includes a description and a pre-written TEM label.

16 THREATS IN 5 CATEGORIES

📢 ATC — 4 threats

- **Diversion clearance** — unexpected radar vector toward alternate airfield
- **Radio communication loss** — NORDO, squawk 7600, continuation decision
- **Unexpected TCAS RA** — "CLIMB CLIMB" without ATC warning, mandatory RA follow
- **Blocked runway** — TWR "Go around, aircraft on runway", immediate go-around

CABIN — 4 threats

- **Passenger medical emergency** — unconscious, defibrillator, diversion decision
- **Unarmed door reported** — 2L on final, confirmation required before landing
- **Cabin smoke** — rear area, unidentified source, diversion?
- **Disruptive passenger** — threatening, squawk 7500 considered, ATC coordination

🌩 WEATHER — 3 threats

- **Alternate unavailable** — sudden fog, RVR below minima, limited fuel
- **Reported windshear** — PIREP windshear 400ft, 20kt loss, go-around?
- **Unexpected icing** — moderate to severe not forecast on ATIS, anti-ice activated

RUNWAY — 1 threat

- **Runway contamination** — Braking action POOR, performance not recalculated, hold or land?

⚙ SYSTEM — 4 threats

- **Fuel imbalance** — ECAM FUEL IMBAL increasing, balance fuel procedure
- **Abnormal pressurisation** — ECAM CABIN ALTITUDE, emergency descent considered?
- **Cargo fire warning** — ECAM FWD CARGO FIRE, NNC cargo fire, immediate diversion
- **Bird strike** — detected during takeoff/approach, engine check, return?

TRI BEST PRACTICE

- Inject **without warning** the crew mid-procedure. Observe: who detects first (SA), who prioritises (WLM), who communicates (COM). Record the reaction immediately in TEM before continuing. The threat is automatically timestamped.

MANDATORY — COLORS & FULL PROCEDURE

The 9 **MANDATORY** EASA items appear in a coloured bar at the top of the exercise list. Each item has its own neon colour for instant identification:

RTO

Rejected Takeoff

EF V1-V2

Engine Failure V1-V2

SE ILS

Single Engine ILS

SE GO-AROUND

Single Engine Go-Around

SE LANDING

Single Engine Landing

NON-ILS

Non-ILS to Minima

TCAS RA

TCAS Resolution Advisory

WINDSHEAR

Windshear Escape Maneuver

UPSET RECOVERY

Unusual Attitude Recovery

HOW TO USE THE MANDATORY BAR

RTO

Dark background, greyed text → exercise **not yet completed**

RTO

Coloured background + → exercise **closed successfully**

🔄 2nd ATTEMPT — PROCEDURE

- 1 The MANDATORY exercise is graded **< 3 (below standard)** and closed
- 2 A **2nd** badge and a 🔄 2nd ATTEMPT button appear in the list
- 3 Press 🔄 2nd ATTEMPT → a duplicate exercise is added to the list
- 4 Redo the exercise → close it → the 2nd attempt is graded separately in the debrief

⚠️ EASA REGULATION

All 9 MANDATORY items must be completed and closed to validate an EASA OPC/PC. One unclosed item = check not valid under regulations. The MANDATORY bar gives you real-time status.

FORDEC

Decision-making model — use it when facing any abnormal situation

F

FACTS

What is happening? Gather and share the available information.

O

OPTIONS

What alternatives exist? List without judging, involve the crew.

R

RISKS & BENEFITS

Risks and benefits of each option. Which one is best?

D

DECISION

Decision made and announced. Crew aligned on the action plan.

E

EXECUTION

Who does what? Role allocation and procedure execution.

C

CHECK

Did the action produce the expected result? Confirm or reconsider.

IN THE APP — FORDEC Tab

Open an exercise → **FORDEC** tab → check **FORDEC USED** if the crew applied the method. A **F** badge appears on the card and the debrief displays the usage rate.

NITS BRIEF

Crew brief — done AFTER FORDEC, BEFORE executing the procedure



NATURE

Nature of the failure or abnormal situation detected.



INTENTION

What the PIC intends to do — the chosen procedure and its objective.



TIME

Time available or execution moment — urgency or time to think?



SPECIAL INSTRUCTIONS

Special instructions, precautions, specific PF/PM roles.

COMPLETE SEQUENCE

Failure detected → **FORDEC** (decide) → **NITS** (brief the crew) → Execute the procedure. A **N** badge appears on the card if the NITS brief was completed.

SOURCES & REFERENCES

OFFICIAL SOURCES — VALIDATED CONTENT

ICAO Doc 9995 — CBTA Manual

Framework for the 9 competencies, definitions, and observable behaviors. Normative basis for all CBTA assessment in this app.

EASA Part-FCL — Appendix 9

Regulatory requirements for OPC, PC and Line Checks. Session types, mandatory exercises and frequencies.

EASA AMC/GM to Part-FCL — CBTA

Grading scale 1 → 5 (UNACCEPTABLE to EXEMPLARY) and competency assessment criteria.

ICAO Doc 9683 — Human Factors Training Manual

TEM framework (Threat and Error Management) — threats, errors, undesired aircraft states.

FORDEC — Lufthansa Aviation Training

Crew decision-making model. CRM standard adopted by the global aviation industry.

NITS — Commercial Aviation CRM Standard

Crew briefing model (Nature · Intention · Time · Special Instructions). Industry standard.

REGULATORY DISCLAIMER

This application is a **TRI support tool** — it does not replace the official Boeing CBTA Package, QRH, FCTM or your operator's OM-D. All B737-specific content must be validated by the TRI against documents approved by the competent authority before operational use. **Responsibility for the assessment remains with the instructor.**

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